Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid

C.-C. Wang, Y.-P. Huang, S. Yip, Phys. Rev. B **114**, 305105

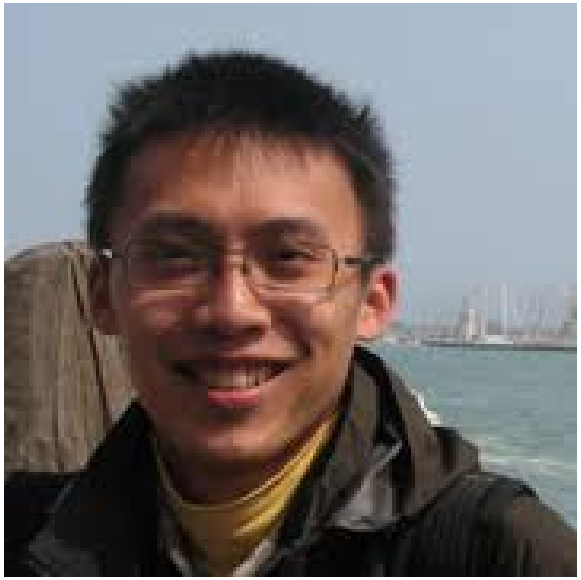
Chen-Chih Wang

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November 18, 2026

Outline

Collaborators



Prof. Yi-Ping Huang
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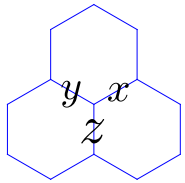


Dr. Sungkit Yip
Academia Sinica

Kitaev Model: an Introduction

- Model Hamiltonian

$$H = -\sum_{\langle ij \rangle} J_{\alpha_{ij}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \quad ; \quad$$



- Majorana fermion (parton) representation: b^x, b^y, b^z, $c \Rightarrow \sigma^\mu = i b^\mu c$
- Parton Hamiltonian: define $i b_i^{\alpha_{ij}} b_j^{\alpha_{ij}} = \hat{u}_{ij}$

$$H \to \sum_{\langle ij \rangle} i J_{\alpha_{ij}} \hat{u}_{ij} c_i c_j$$

- Gutzwiller projection

$$\mathcal{P}_G = \prod_i \left(\frac{\mathbb{1} + D_i}{2} \right) \quad , \quad D_i = b_i^x b_i^y b_i^z c_i \Leftarrow -i \sigma_i^x \sigma_i^y \sigma_i^z$$



Kitaev Model: Gauge Theory

In the Majorana fermion representation, the model is equivalent to a system of free Majorana fermions (matter field) coupled to a static $\mathbb{Z}_2$ gauge field

- Define the **gauge field** $\hat{u}_{ij} = e^{i\boldsymbol{A_{ij}}}$, then the elementary $\mathbb{Z}_2$ **gauge flux** reads

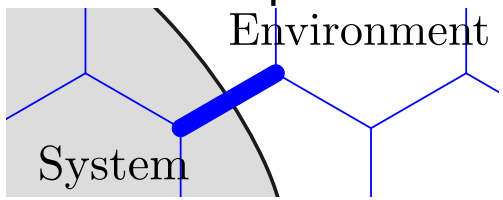
$$e^{i\Phi} = \exp\left(i \oint_{\partial p} A\right) = \prod_{\langle ij\rangle \in \partial p} \hat{u}_{ij} = \prod_{\langle ij\rangle \in \partial p} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} =: W_p$$

- The **electric field** in the gauge field that flips the gauge field is $\boxed{b_i^{\alpha_{ij}} \to e^{i\pi \boldsymbol{E_{ij}}}}$ because $b_i^{\alpha_{ij}} \hat{u}_{ij} b_i^{\alpha_{ij}} = -\hat{u}_{ij} \Longrightarrow [\boldsymbol{A_{ij}}, \boldsymbol{E_{i'j'}}] = i\delta_{ii'}\delta_{jj'}$
- **Gutzwiller projection** as a **Gauss law** constraint: $D_i = b_i^x b_i^y b_i^z c_i = 1$. The ground state is

$$\prod_i \left(\frac{\mathbb{1} + D_i}{2}\right) |u\rangle |M\rangle \quad ; \quad \text{Fix the gauge: } \langle u| \hat{u}_{ij} |u\rangle = +1 \forall \langle ij\rangle$$

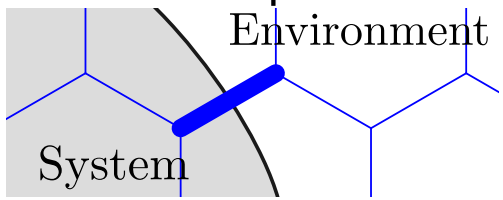
Motivation

- A direct **entanglement** analysis from the **physical density matrix**
  - Gutzwiller projection is needed in the fermionic representation.
  - Using physical **spin**s rather than Majorana fermions
  - Odd number of sites: ill-defined Majorana fermionic Hilbert space
  - Cutting gauge fields on links (add figures)



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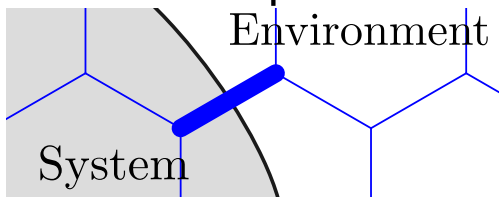


- $\mathbb{Z}_2$ **gauge theory** with **1-form symmetry**: organization of entanglement spectrum
  - How do these symmetries restrict the entanglement spectrum structure?
  - Directly solving the entanglement spectrum rather than using the replica trick

(Xu *et al.* PRX **15**, 011001 (2025))

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Recent New Development:

- Subsystems with holes (genus number $g > 0$)
- Quantum magic: direct calculation of the *stabilizer Rényi entropy*

Kitaev Model: Properties of Spin Correlation Functions

- Two-spin correlation functions are short-ranged (Baskaran, Mandal, and Shankar, PRL**98**, 247201)

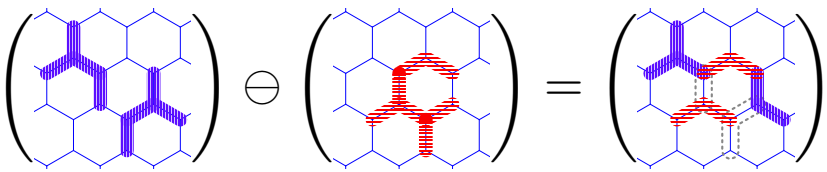
$$\left\langle \sigma_i^\mu \sigma_j^\nu \right\rangle = \langle M | \, i c_i c_j \, | M \rangle \qquad \underbrace{\langle u | \, i b_i^\mu b_j^\nu \, | u \rangle}_{\text{vanishes if } i b_i^\mu b_j^\nu \neq \hat{u}_{ij}}$$

- Multi-spin correlation functions can ONLY be string-like

$$\left\langle \prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle \equiv \langle \Sigma_{\mathfrak{D}} \rangle$$

- $\mathbb{Z}_2$ algebra of strings

$$\mathfrak{D} \ominus \mathfrak{D}' := \mathfrak{D} \cup \mathfrak{D}' - \mathfrak{D} \cap \mathfrak{D}' \quad : \quad$$



- String operators form a projective representation

$$\boxed{\Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}'} = \varphi(\mathfrak{D}, \mathfrak{D}') \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'}} \quad ; \quad \varphi(\mathfrak{D}, \mathfrak{D}') = \pm 1$$

Organization: Equivalence Relation

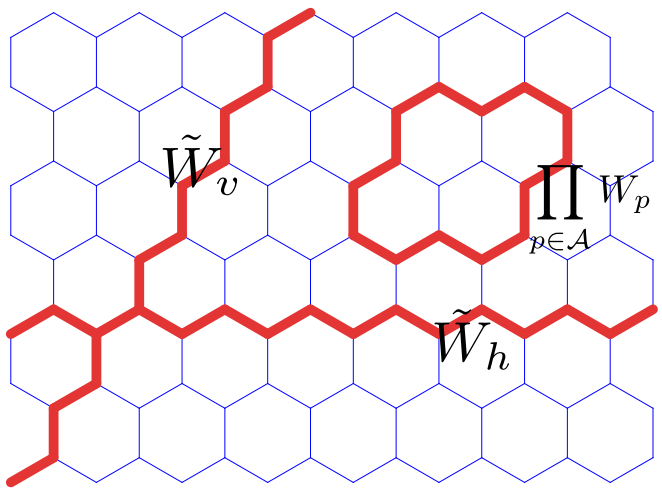
- Exact 1-form symmetry: Wilson loop $\Sigma_{\mathcal{C}} |\psi_0\rangle = W_{\mathcal{C}} |\psi_0\rangle = |\psi_0\rangle$
- The expectation values are identical up to a sign if the strings are identical up to a loop $\mathcal{C}$

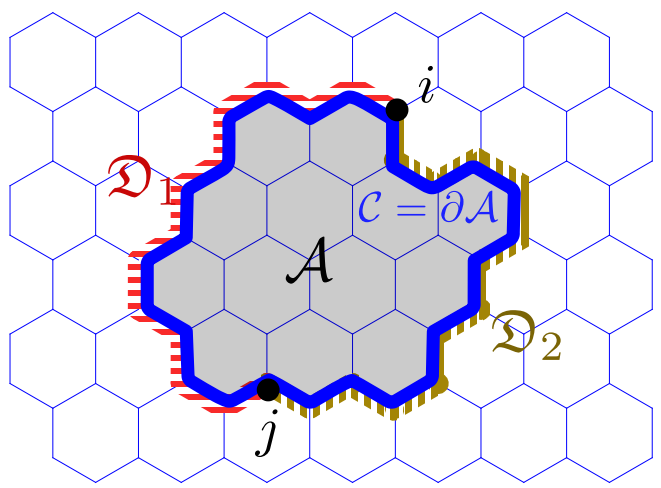
$$\langle \Sigma_{\mathfrak{D} \ominus \mathcal{C}} \rangle = \varphi(\mathfrak{D}, \mathcal{C}) \langle \Sigma_{\mathfrak{D}} \Sigma_{\mathcal{C}} \rangle = \varphi(\mathfrak{D}, \mathcal{C}) \langle \Sigma_{\mathfrak{D}} W_{\mathcal{C}} \rangle = \varphi(\mathfrak{D}, \partial A) \langle \Sigma_{\mathfrak{D}} \rangle$$

- Equivalence relation:

$$\boxed{[\mathfrak{D}] = \left\{ \mathfrak{D}' \ : \ \exists\, \mathcal{C} \text{ composed by closed loops such that } \mathfrak{D}' \ominus \mathcal{C} = \mathfrak{D} \right\}}$$

- Endpoints perspective: $\mathrm{EP}(\mathfrak{D}) = \mathrm{EP}(\mathfrak{D}')$ iff $[\mathfrak{D}] = [\mathfrak{D}'] \Rightarrow$ Emerging fermions at endpoints of strings





The 1st Layer Organization

- The density matrix of a spin-1/2 system can be expanded in terms of Pauli-group elements

$$\rho \propto \sum_{\mathsf{S}\in\mathcal{P}_N} \langle \mathsf{S} \rangle \, \mathsf{S}$$

- The string operators $\Sigma_{\mathfrak{D}}$ in the spin-$1/2$ Kitaev model are Pauli strings up to a sign.

$$\rho \propto \sum_{\mathfrak{D}} \underbrace{\eta_{\mathfrak{D}} \langle \Sigma_{\mathfrak{D}} \rangle}_{\mathfrak{e}_{\mathfrak{D}}} \Sigma_{\mathfrak{D}} \equiv \sum_{\mathfrak{D}} \mathfrak{e}_{\mathfrak{D}} \Sigma_{\mathfrak{D}}$$

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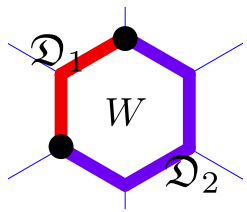
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- Strings belonging to the same class can be grouped.

$$\mathfrak{C}_{\mathfrak{D}_1} \Sigma_{\mathfrak{D}_1} + \mathfrak{C}_{\mathfrak{D}_2} \Sigma_{\mathfrak{D}_2} \to \mathfrak{C}_{\mathfrak{D}_1} \Sigma_{\mathfrak{D}_1} + \mathfrak{C}_{\mathfrak{D}_1} W \Sigma_{\mathfrak{D}_1}$$



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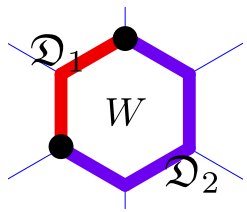
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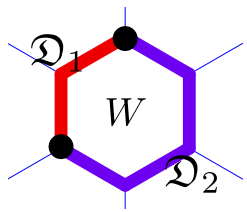
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- Organization due to the exact 1-form Wilson symmetry:

$$\rho \propto \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}} \hat{\Pi} \Sigma_{\mathfrak{D}} \quad ; \quad \hat{\Pi} := \underbrace{\left(\frac{\mathbb{1} + \tilde{W}_h}{2}\right)\left(\frac{\mathbb{1} + \tilde{W}_v}{2}\right)}_{\text{topological projector (global Wilson)}} \underbrace{\prod_p \left(\frac{\mathbb{1} + W_p}{2}\right)}_{\text{flux-free projector}}$$

The 1st Layer Organization

Normalize $\mathrm{Tr}\rho = 1$ to get the final result:

$$\boxed{\rho = \frac{1}{2^{N/2}} \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}} \hat{\Pi} \Sigma_{\mathfrak{D}}}$$

Physical Interpretation:

Since $\mathrm{EP}(\mathfrak{D}) = \mathrm{EP}(\mathfrak{D}')$ iff $[\mathfrak{D}] = [\mathfrak{D}']$, the equivalent fermion density matrix ρ^F looks the same as ρ.

$$\rho^F = \frac{1}{2^{N/2}} \sum_{\mathrm{EP}(\mathfrak{D})} (\text{fermion Op.}) = \frac{1}{2^{N/2}} \sum_{[\mathfrak{D}] \subset A} (\text{fermion Op.})$$

The 1st Layer Organization

- Following the same spirit to construct the reduced density matrix:

$$\rho_A \propto \sum_{\mathfrak{D}\subset A} \mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}} \xrightarrow{\text{normalize}} \boxed{\frac{1}{2^{N^A-N_p^A}} \sum_{[\mathfrak{D}]\subset A} \mathfrak{C}_{\mathfrak{D}}\hat{\Pi}_A\Sigma_{\mathfrak{D}}}$$

- A geometric relation to area law: $\boxed{|\partial A| = N^A - 2N_p^A + 2}$
- Physical interpretation: **area law** + **Majorana fermion**

$$\boxed{\rho_A = \frac{1}{2^{(|\partial A|/2-1)}} \left(\frac{1}{2^{N^A/2}} \sum_{[\mathfrak{D}]\subset A} \mathfrak{C}_{\mathfrak{D}}\hat{\Pi}_A\Sigma_{\mathfrak{D}} \right)}$$

The equivalent fermion density matrix ρ_A^F looks the same as ρ_A.

$$\rho_A^F = \frac{1}{2^{N^A/2}} \sum_{\mathrm{EP}(\mathfrak{D})\subset A} (\text{fermion Op.}) = \frac{1}{2^{N^A/2}} \sum_{[\mathfrak{D}]\subset A} (\text{fermion Op.})$$

Organization: Block-Diagonal Structure

- *Majorana representation* of string operators: $\mathcal{M}_{\mathfrak{D}} := \prod_{\langle ij\rangle \in \mathfrak{D}} ic_j c_i$, which satisfies the same algebra (statistics)

$$\mathcal{M}_{\mathfrak{D}} \mathcal{M}_{\mathfrak{D}'} = \varphi(\mathfrak{D}, \mathfrak{D}') \mathcal{M}_{\mathfrak{D} \ominus \mathfrak{D}'}$$

- Difference in the Hilbert space dimension:

$$\dim \mathcal{H}_S / \dim \mathcal{H}_F = 2^{|\partial A|/2-1}$$

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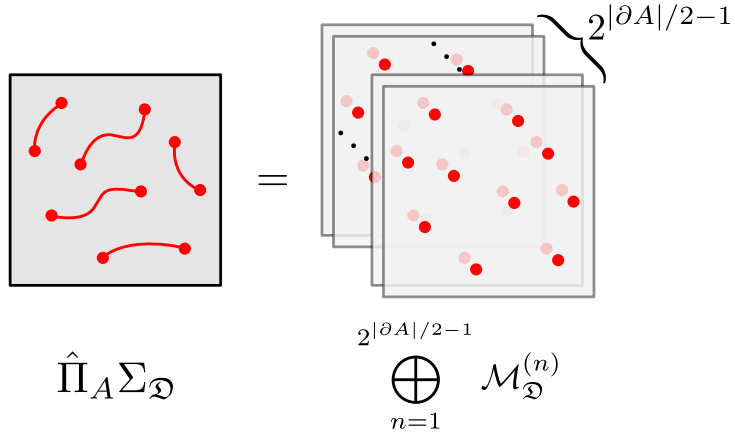
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- Conjecture: projected string operators are **block diagonal** $\rightarrow$ Extensive degeneracy

$$\hat{\Pi}\Sigma_{\mathfrak{D}} = \frac{1}{2^{|\partial A|/2-1}} \bigoplus_{n=1}^{2^{|\partial A|/2-1}} \mathcal{M}_{\mathfrak{D}}^{(n)}$$



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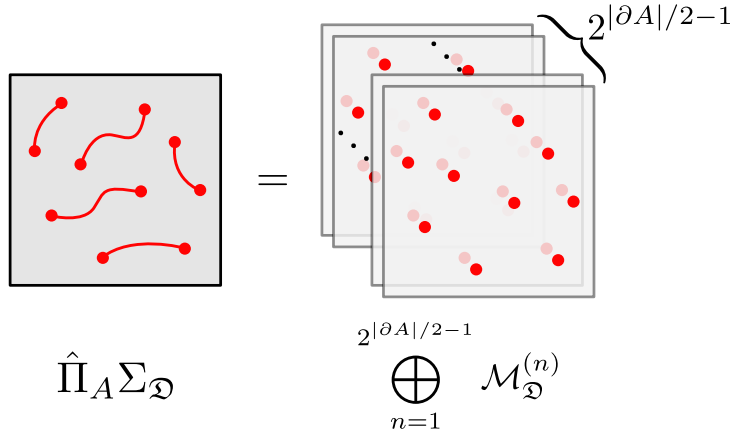
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Observation: A degenerate entanglement spectrum leads to the separability of the entanglement entropy

$$\rho = \frac{1}{d} \bigoplus_{n=1}^{d} \varrho \Rightarrow R_n(\rho) = \frac{1}{1-n} \ln \left[\frac{1}{d^{n-1}} \mathrm{Tr}\left(\varrho^n\right) \right] = \underbrace{R_n(\varrho)}_{\text{matter}} + \underbrace{\ln d}_{\text{gauge}}$$

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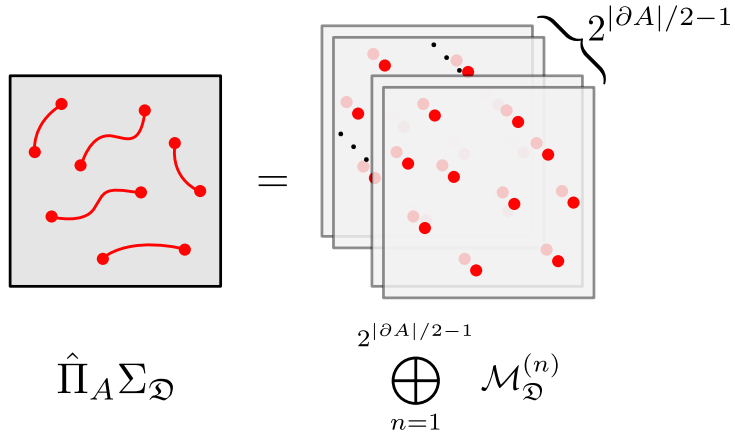
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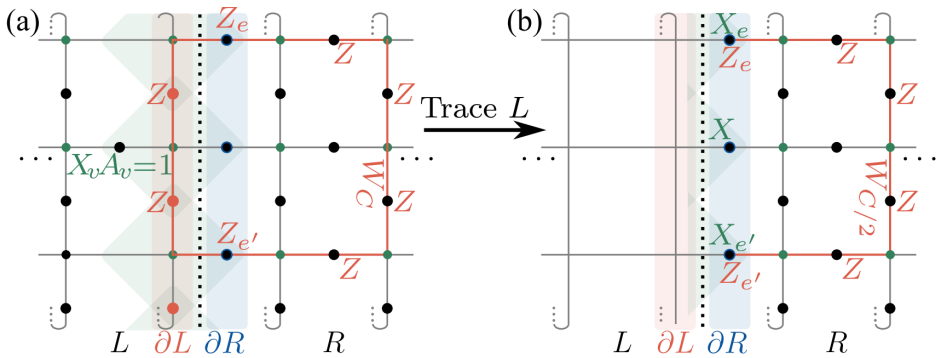
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Questions:

- Can we directly prove that the entanglement spectrum is degenerate?
- How could they be related to *gauge fields*?

Exact Gauss Law × Exact 1-Form Wilson Symmetry: –Exact Block-Diagonal Structure

Review the method discovered by a recent work [W.-T. Xu *et al.* PRX **15**, 011001 (2025)]



- The Gauss law operator in a $\mathbb{Z}_2$ gauge theory: $X_v \prod_{e\in\mathcal{N}(v)} X_e \equiv X_v A_v$
- Exact Gauss law $[\rho, X_e] = 0\ \forall e \in \partial R$: block-diagonal structure $\rho = \bigoplus_{\{x\}} \varrho_{\{x\}}$
- Exact 1-form Wilson loop symmetry $[W_{\mathcal{C}/2}, \rho] = 0$ relates different blocks since $W_{\mathcal{C}/2} X_e W_{\mathcal{C}/2}^\dagger = -X_e\ \forall e \in \partial R$
- Yielding $2^{|\partial R|-1}$-fold degeneracy $\rho = \frac{1}{2^{|\partial R|-1}} \varrho^{\oplus 2^{|\partial R|-1}}$

Block-Diagonal Density Matrix in the Kitaev Model

In the Kitaev model:

- Exact Gauss law: From the electric field $b_i^{\alpha_{ij}} = e^{i\pi E_{ij}}$ at the boundary.
- Exact 1-form Wilson symmetry: $\langle W_{\mathcal{C}} \rangle = \langle \Sigma_{\mathcal{C}} \rangle = 1$ for any closed loop $\mathcal{C}$.

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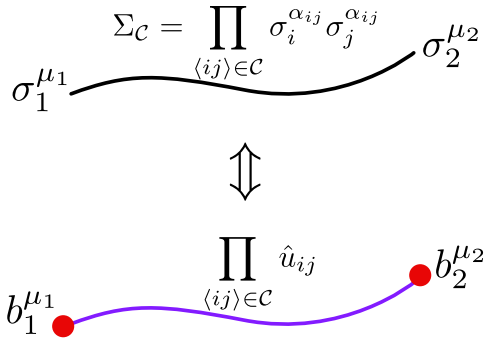
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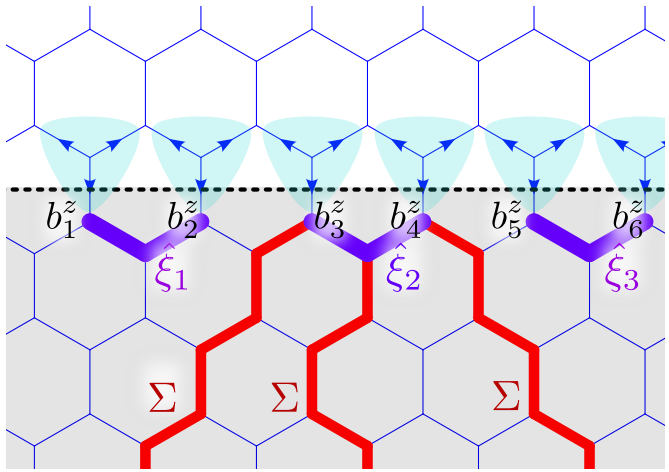
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But the gauge field here is *fictitious*. To use the spin representation (physical):

- Connecting two *deconfined* degrees of freedom to form physical objects
- Use electric fields to define a set of operators $\hat{\xi}_n := \sigma_n^{\mu_n} \Sigma_{\mathcal{C}_n} \sigma_{n+1}^{\mu_{n+1}}$ along the boundary of the entanglement cut. These operators commute with strings in A:

$$\boxed{[\Sigma_{\mathfrak{D}}, \hat{\xi}_n] = 0 \quad ; \quad \mathfrak{D} \subset A}$$

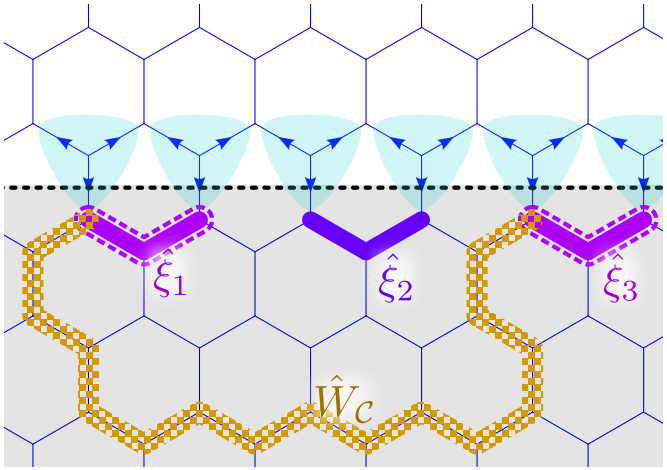


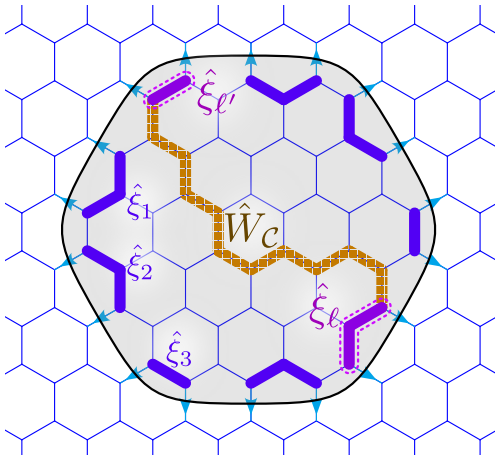


The 2nd Layer Organization

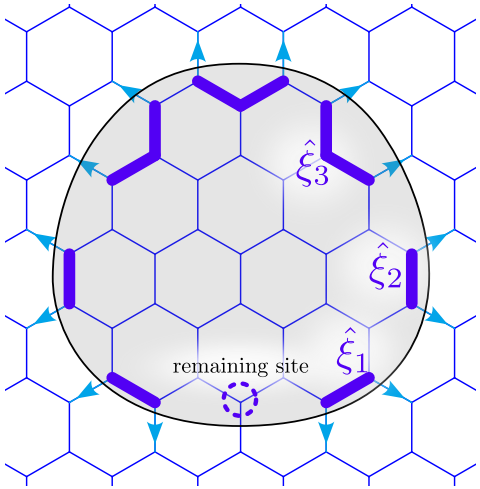
- There are $L = \frac{1}{2}|\partial A|$ independent $\hat{\xi}$-operators along the boundary.
- String operators are block-diagonalized by $\hat{\xi}$-operators
- Non-contractible Wilson operators flip $\hat{\xi}$-operators, making different blocks identical
- The reduced density matrix is also block-diagonal: extensive degeneracy of the entanglement spectrum

$$\boxed{\rho_A = \frac{1}{2^{L-1}} \bigoplus_{n=1}^{2^{L-1}} \rho_A^F} \Rightarrow S_A^{(n)} = \frac{1}{1-n} \ln \mathrm{Tr} \rho_A^n = \underbrace{S^{(n)}(\rho_A^F)}_{\text{matter}} + \underbrace{\left(\frac{1}{2}|\partial A| - 1\right) \ln 2}_{\text{gauge}}$$



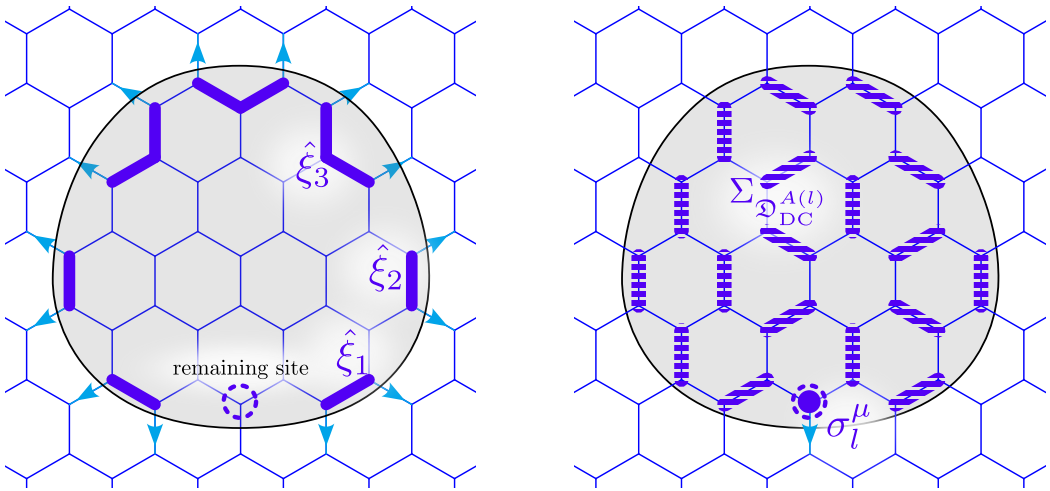


Odd Number of Sites



- When N^A is odd, an equivalent fermion system is ill-defined: We need to borrow a fermion from the gauge sector.
- For odd N^A, there is always an unpaired site at the boundary.

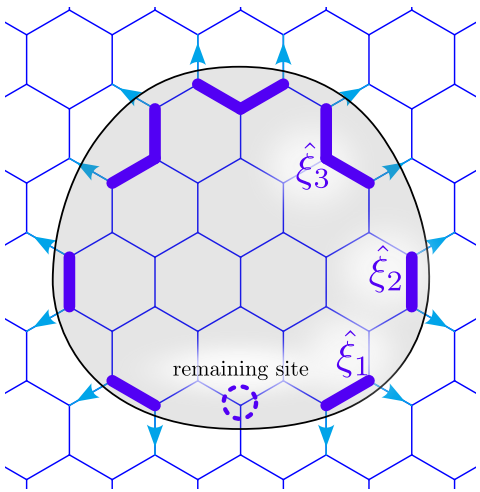
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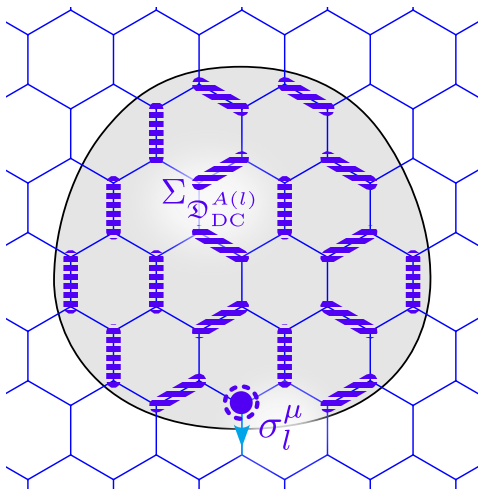


- When N^A is odd, an equivalent fermion system is ill-defined: We need to borrow a fermion from the gauge sector.
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- We can define an additional operator Ξ_A that also commutes with ρ_A, yielding an additional 2-fold degeneracy in the entanglement spectrum

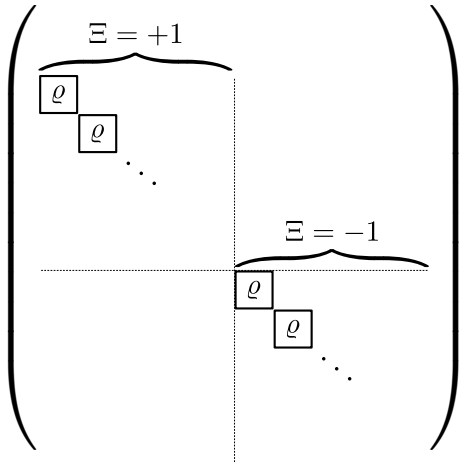
$$\hat{\Xi}_A \equiv \sigma_l^\mu \Sigma_{\mathfrak{D}_{\rm DC}^{A(l)}} \to \left(\prod_{\langle ij \rangle \in \mathfrak{D}_{\rm DC}^{A(l)}} \hat{u}_{ij} \right) \underbrace{\left(b_l^\mu \prod_{i \in A} c_i \right)}_{\text{fermion parity}}$$

Odd Number of Sites





$$\rho_A \cong$$



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Quantum Magic

- The plaquette operators and global loops form a stabilizer group $\langle W_{p_1}, W_{p_2}, \cdots \rangle$
- The **gauge part** of the Kitaev ground state ($|\mathrm{Gnd}\rangle \propto \rho\,|\phi\rangle$) is a **stabilizer projector**.

$$|\mathrm{Gnd}\rangle = C(\phi) \underbrace{\left(\frac{\mathbb{1}+\tilde{W}_v}{2}\right)\left(\frac{\mathbb{1}+\tilde{W}_h}{2}\right)\prod_p\left(\frac{\mathbb{1}+W_p}{2}\right)}_{=\hat{\Pi}\quad \text{gauge (=stabilizer)}} \underbrace{\left[\frac{1}{2^{N-N_p}}\sum_{[\mathfrak{D}]}\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}}\right]}_{\text{matter}} |\phi\rangle$$

- The **matter part** controls the **non-stabilizerness** of the state.

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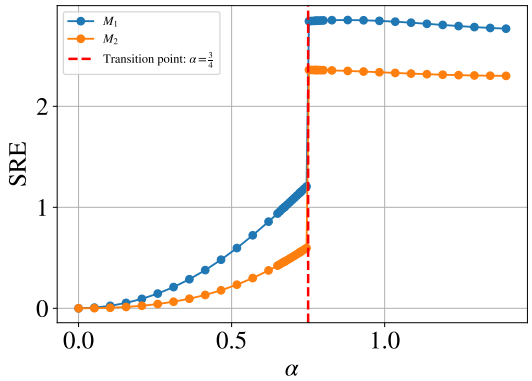
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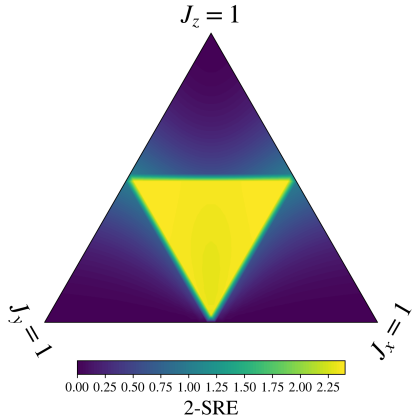
Stabilizer Rényi Entropy (SRE)

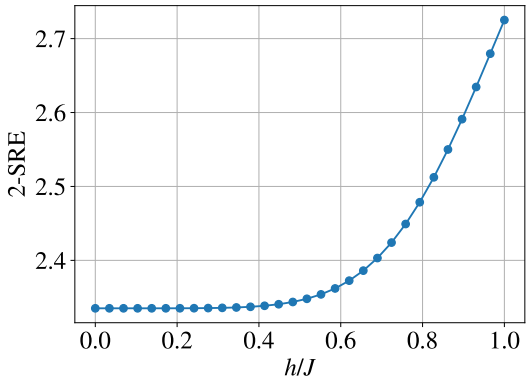
$$M_n\left(|\psi\rangle\right) := \frac{1}{1-n}\ln\left(\sum_{\mathsf{S}\in\mathcal{P}_N}\frac{1}{2^N}\mathrm{Tr}\left(\rho\mathsf{S}\right)^{2n}\right) = \frac{1}{1-n}\ln\left(\frac{1}{2^{N/2}}\sum_{[\mathfrak{D}]}\eta_{\mathfrak{D}}^n\mathfrak{C}_{\mathfrak{D}}^{2n}\right)$$

- The result depends only on the equivalence classes (endpoints), and the number of effective degrees of freedom is $N/2$.

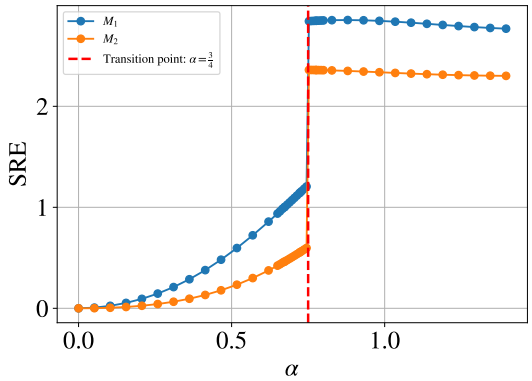
Quantum Magic

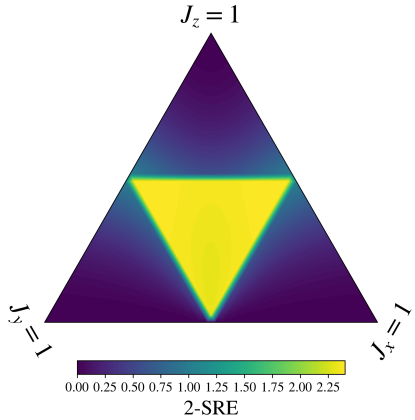


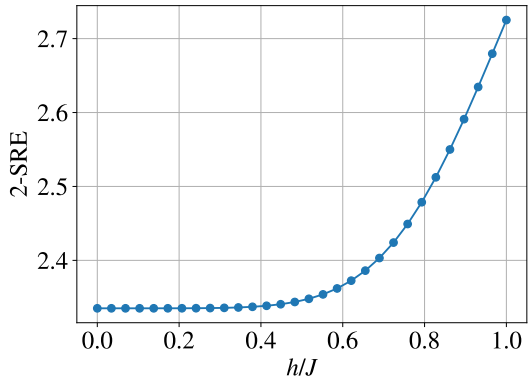




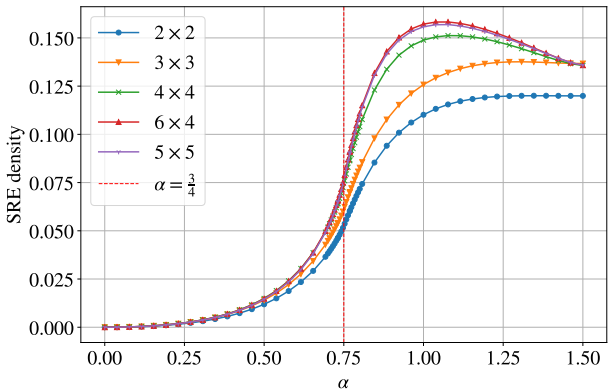
Quantum Magic

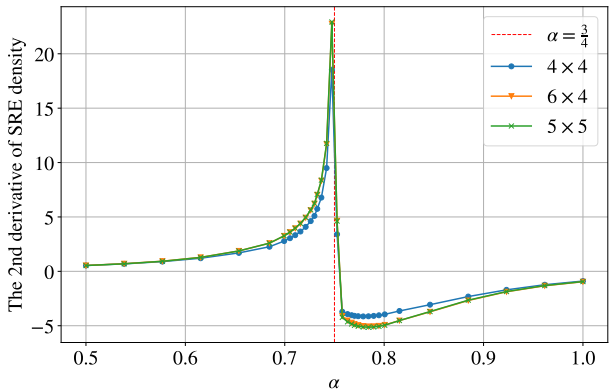


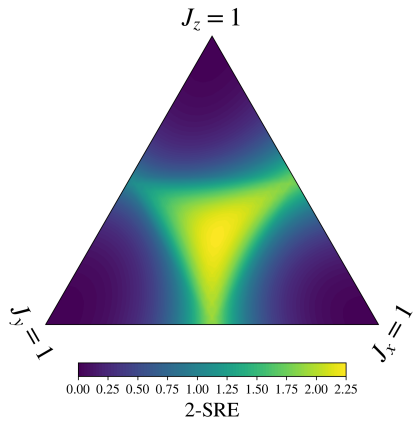




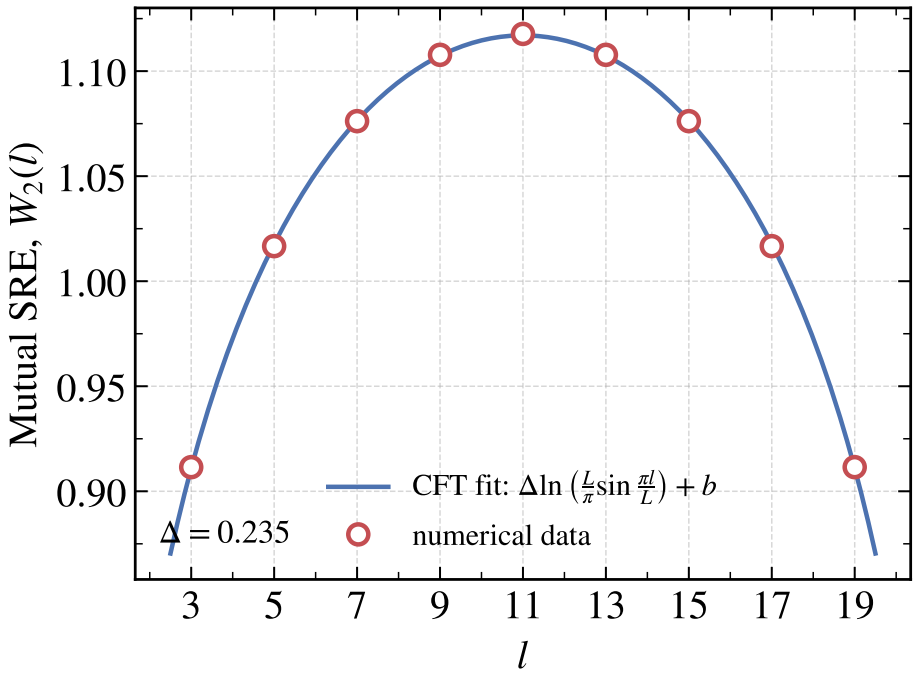
Mixed-State SRE: $\tilde{M}_2(\rho) = -\ln\left(\frac{1}{2^N}\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_N}\mathrm{Tr}\left(\rho\mathsf{S}\right)^4\right) - R_2(\rho)$







Quantum Magic



Summary

- The entanglement structure is demonstrated in terms of **physical spins**.
- The exact **1-form Wilson symmetry** makes the density matrix **organized**.
- The **exact 1-form Wilson symmetry** with the **exact Gauss law** gives an extensively degenerate entanglement spectrum.
  - *Glue* the fictitious electric fields back to form physical operators $\hat{\xi}$ in the spin representation.
  - The reduced density matrix is composed of identical copies of the fermionic density matrix.
  - Separability of the entanglement entropy
- **Odd** number of lattice sites: **Fermion parity** provides an additional degeneracy.
- The **gauge part** is referred to as the **stabilizer**, while the **matter part** governs the **non-stabilizerness**.

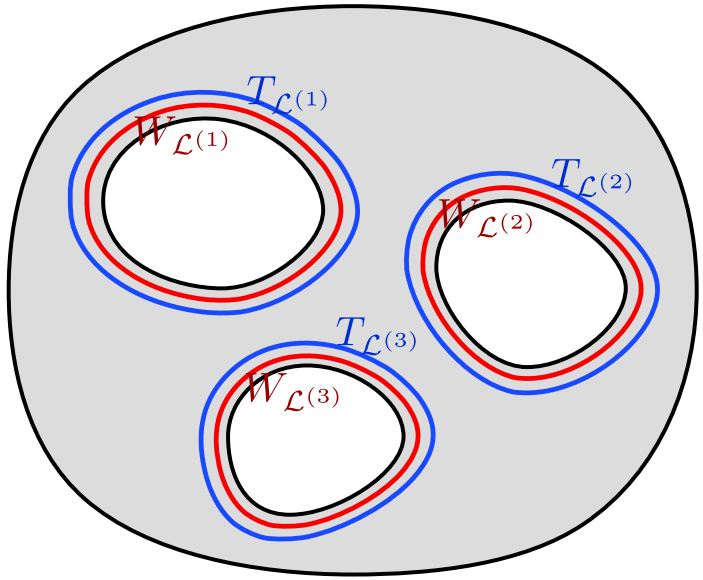
Unsolved Problems and Outlook

- To get the **true topological entanglement entropy**, we need an exact 't Hooft symmetry and make it spontaneously broken.
- Higher spin Kitaev model: Expectation values don't provide sufficient conditions.
- Multi-connected subregion

Toric Code Model: True Topological Entanglement entropy

- Reduced density matrix of a Kitaev ground state

$$\rho_A = \frac{1}{2^{N^A - N_p^A - N_s^A - 2g}} \prod_{n=1}^{g} \left(\frac{\mathbb{1} + T_{\mathcal{L}^{(n)}}}{2} \right) \prod_{n=1}^{g} \left(\frac{\mathbb{1} + W_{\mathcal{L}^{(n)}}}{2} \right) \prod_{p \in A} \left(\frac{\mathbb{1} + P_p}{2} \right) \prod_{s \in A} \left(\frac{\mathbb{1} + S_s}{2} \right)$$

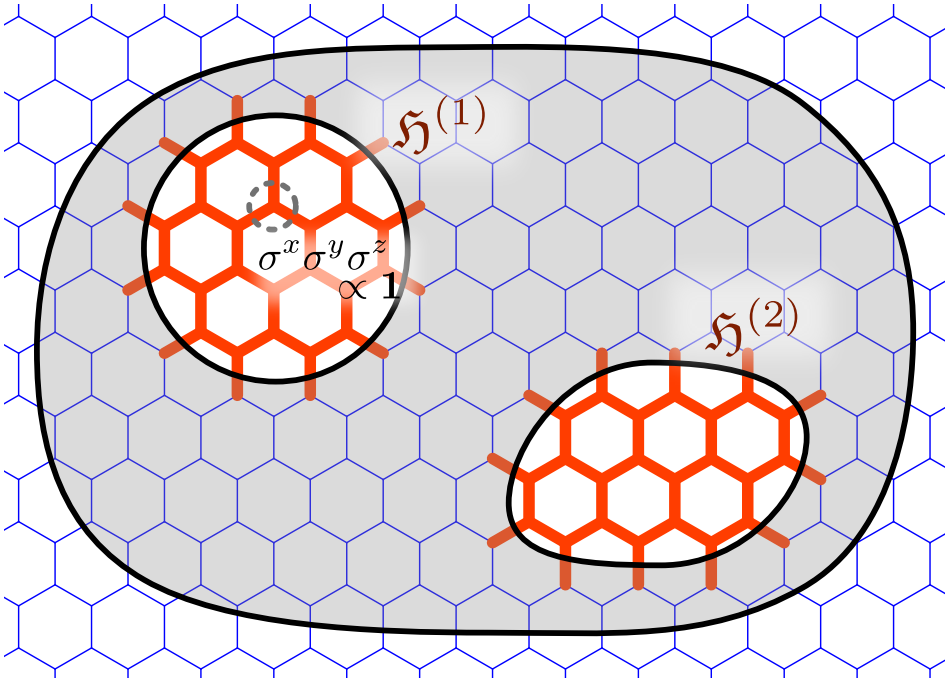


- Entanglement entropy: If the density matrix is proportional to a projector $\rho_A = C\mathcal{P}$

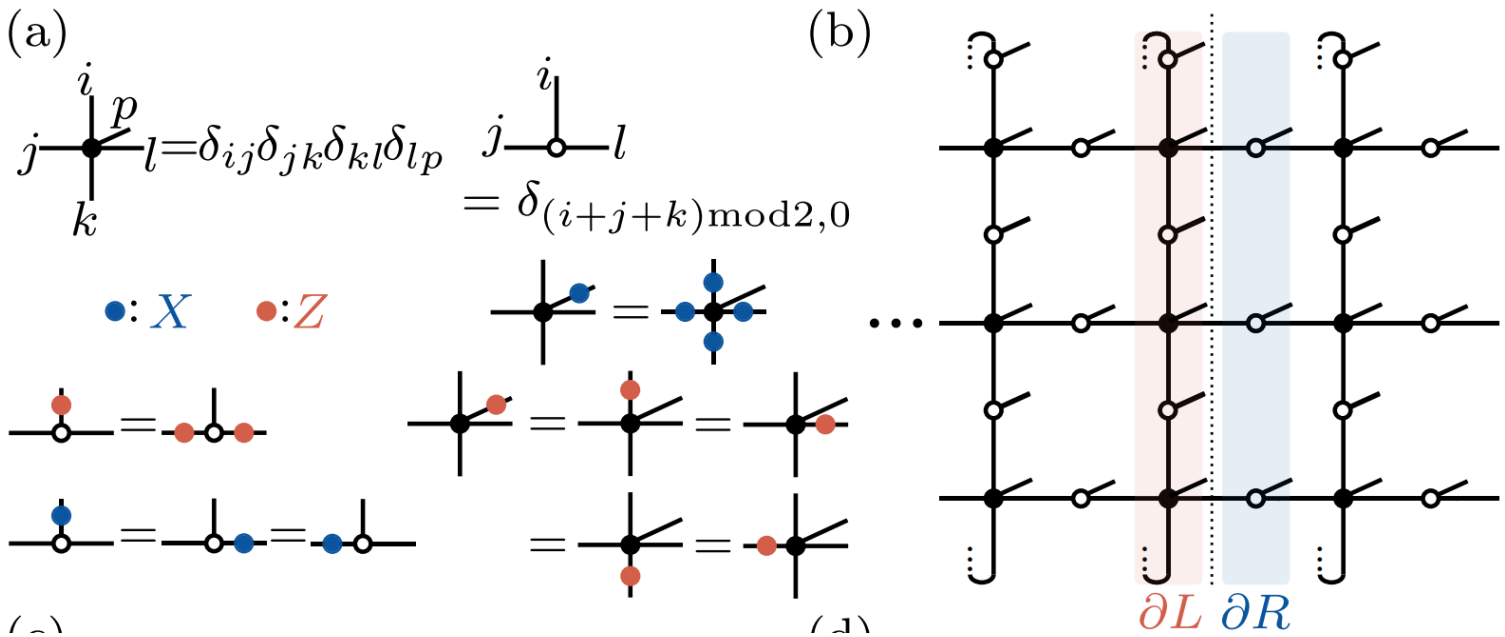
$$S_A^{(n)} = \frac{1}{1-n} \ln \mathrm{Tr} \rho_A^n = \frac{1}{1-n} \ln \left(C^{n-1} \mathrm{Tr} \rho_A \right) = -\ln C = (|\partial A| - (j+g)) \ln 2$$

Multi-connected Region

- For a multi-connected region with genus number g, there is a set of configurations $\left\{\mathfrak{H}^{(1)}, \mathfrak{H}^{(2)}, \cdots, \mathfrak{H}^{(g)}\right\}$ such that $\mathrm{Tr}_{\bar{A}}\left(\Sigma_{\mathfrak{D}\ominus\mathfrak{H}^{(n)}}\right) \neq 0$ if $\mathfrak{D} \subset A$
- In general, however, $\Sigma_{\mathfrak{D}\ominus\mathfrak{H}^{(n)}} \neq \Sigma_{\mathfrak{D}}\Sigma_{\mathfrak{H}^{(n)}}$, thus the reduced density matrix cannot be organized as before.
- The web patch is invalid in the fermion representation: $\mathrm{Tr}_{\bar{A},F}\left(\mathcal{M}_{\mathfrak{D}\ominus\mathfrak{H}^{(n)}}\right) = 0$

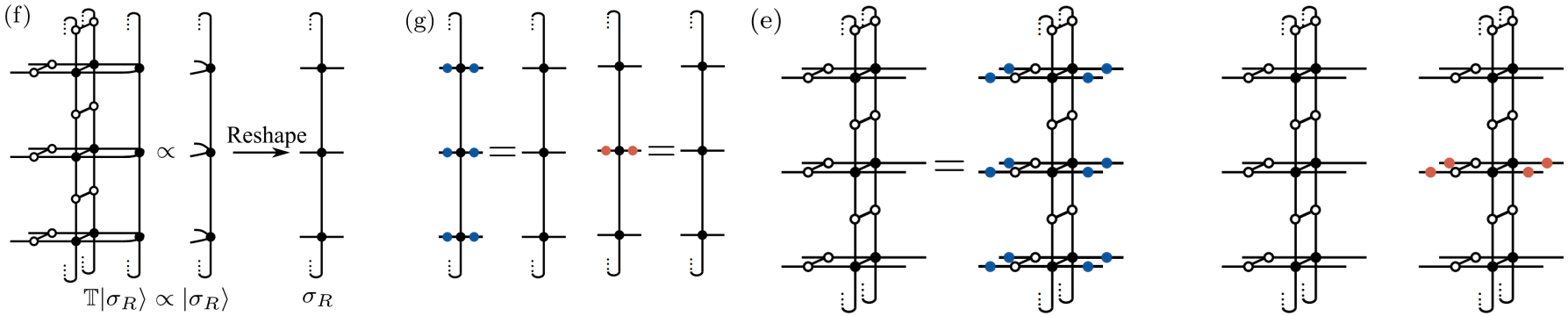


Contribution 2: Exact 1-Form Wilson



- **Symmetry of the tensor network** derives the **0-form Ising symmetry** and the **1-form Wilson loop symmetry**.

Contribution 2: Exact 1-Form Wilson



- Transfer matrix of a tensor network gives reduced density matrix (in the virtual space)

$$\mathbb{T}\,|\sigma_R\rangle \propto |\sigma_R\rangle \quad \Rightarrow \quad |\sigma_R\rangle \to \sigma_L \quad \Rightarrow \quad \rho \sim \sigma_L^T \sigma_R$$

- Symmetry of the tensor network becomes the symmetry of the reduced density matrix

$$\left(\prod_i X_i \otimes \prod_i X_i\right) \mathbb{T} \left(\prod_i X_i \otimes \prod_i X_i\right) = \mathbb{T} \quad \Rightarrow \quad \left(\prod_i X_i\right) \sigma_{L(R)} \left(\prod_i X_i\right) = \sigma_{L(R)}$$

$$(Z_j \otimes Z_j)\mathbb{T}(Z_j \otimes Z_j) = \mathbb{T}\ \forall j \qquad Z_j \sigma_{L(R)} Z_j = \sigma_{L(R)}\ \forall j$$

- Therefore, two operators $\prod_i X_i$ and Z_j commute with the reduced density matrix, yielding 2-fold degeneracy in the entanglement spectrum.

Contribution 3: Spontaneous 1-form Symmetry Breaking

- Implementing the 1-form 't Hooft loop symmetry

$$\rho \sim \frac{1 + T_{C_y}}{2} \exp\left(-H_E\right)$$

- The 1-form 't Hooft loop symmetry is spontaneously broken: $T_{C_y} |\psi\rangle \neq |\psi\rangle$, so the expectation value satisfies the perimeter law: $\langle T_C \rangle \sim e^{-|C|}$.
- Entanglement entropy:

$$S_n = \frac{1}{1-n} \ln \mathrm{Tr} \rho^n \overset{|\partial R| \to \infty}{\approx} -\frac{f(1/n) - f(1)}{1/n - 1} |\partial R| \underbrace{- \ln 2}_{\text{topo EE}}$$

This Wilson is not that Wilson

A Wilson operator defined in the Kitaev model decomposes into a Wilson loop and a 't Hooft loop defined in the toric code (Wen plaquette) model: $W^{(\mathrm{K})} = W^{(\mathrm{t.c.})} T^{(\mathrm{t.c.})}$

